

Rakshita Vijay

The best systems are the ones people never have to think about.

[Chennai, India](#) | [GitHub](#) | [LinkedIn](#)

About

I'm a third-year Computer Science Engineering student at BITS Pilani, Hyderabad, with a habit of picking projects apart down to the data structure underneath.

I've interned at Microsoft and CultureVo, and built independent projects spanning graph neural networks, multi-agent AI systems, and full-stack apps. These experiences taught me how to build systems that hold up under real constraints - correctness, scale, and users who depend on them working.

I served as my school's valedictorian, which fostered a habit of leading from the front early on. That's carried into college, where I've been an active member of my department and taken on organizing roles across multiple clubs and committees.

Outside of internships, I love playing table tennis and cycling, working on a crochet pattern, or getting lost in a mystery or fantasy novel. Right now, I'm focused on deepening my grasp of the systems underneath the AI layer - data pipelines, backend infrastructure, distributed systems - and I'm looking for SDE and applied ML roles where I can keep taking things apart to understand how they hold together.

Education

BITS Pilani, Hyderabad Campus

2023 - 2027

B.E. Computer Science Engineering · CGPA 8.64

St. John's Public School

Class XII · 2023

CBSE · 94.2% · Valedictorian

Chettinad Vidyashram

Class X · 2021

CBSE · 95.8%

Technical Skills

Python, C++, C, Java, SQL, PyTorch, Generative AI, Next.js, React, Supabase, Vercel, Gemini API, Prompt Engineering, Git

Experience

Microsoft - Azure Data Migration Team (Python, PostgreSQL, SQL Server, Prompt Engineering)

May 2026 - Jul 2026

Software Engineering Intern · SQL Server → PostgreSQL Schema Conversion Engine

- Extended an AI-assisted Oracle→PostgreSQL schema-conversion pipeline to also support SQL Server as a source, with a near real-time ETL pipeline.
- Implemented dialect-aware gating across all 4 stages (Extract→Chunk→Convert→Verify), preserving Oracle behavior and aligning with the PG team.
- Built a dependency-aware chunk scheduler on Kahn's Algorithm that topologically orders database objects for conversion - discovering in-scope objects, emitting raw DDL plus a dependency edge graph, and correctly handling cycles and self-referencing edges while upholding ACID transactionality.
- Designed a hybrid conversion router that sent mechanical DDL through a deterministic regex fast-path and reserved LLM-based translation for procedural/semantic logic, using few-shot examples and a shared canonical identifier map to keep naming consistent across both paths - optimizing token usage.
- Built a fault-tolerant compile-and-retry loop that test-compiles each chunk's DDL, defers cross-chunk dependencies, and auto-retries failures from compile-error feedback - ~97% of chunks converged without manual intervention.
- Stress-tested end-to-end against real-world corpora (Northwind, AdventureWorks, WideWorldImporters) plus a custom 7-schema database: ~98% of in-scope objects converted across 45+ chunks (600+ objects with dependencies), cutting runtime from ~4 hours to 35-45 minutes (~83% reduction), correctly identifying and excluding known out-of-scope objects via the fix-loop.

CultureVo (Multi-Agent, Persona System, Multilingual)

May 2025 - Jul 2025

Summer Intern · Conversational AI Tools

- Built PersonaBot, a configurable multi-agent orchestration toolkit with a state-managed persona system spanning 6 profiles and 10 customizable traits.
- Implemented persistent, multi-session conversation state across four languages (English, Sinhala, Tamil, Hindi), enabling bulk generation of 100+ Q&A pairs.

Projects

Android Malware Detection with GNNs (GraphSAGE, Androguard, PyTorch, Skip-Gram)

Jan 2026 - May 2026

Cybersecurity · Graph ML

- Built an end-to-end malware classification pipeline extracting Function Call Graphs (FCGs) via Androguard and training GraphSAGE models to distinguish benign from malicious Android apps.
- Designed an API-aware, structure-preserving graph pruning technique that removes low-signal non-API methods via weighted edge rewiring.
- Integrated pretrained Skip-Gram node embeddings for graph preprocessing/serialization on HPC, with checkpoint-based training for resumable runs.
- Achieved a 94.39% F1-score and 92.41% accuracy on an imbalanced dataset of 150K benign and 50K malicious samples.

Finance Tracker & Analyser (Next.js, Gemini 3.5 Flash, Vercel, Multi-Agent)

Jun 2025 - Jul 2025

AI-Driven Financial Analytics

- Designed a multi-format analysis and report-generation service using Gemini 3.5 Flash, backed by a 2-agent pipeline for automated insight generation.
- Implemented real-time budget tracking with configurable spending categories, plus an AI-driven module for recovery-strategy suggestions.
- Built a rules-based anomaly-detection service that processes uploaded statement data and flags inconsistencies for user review.

Automated Research & Ideation Pipeline (Python, asyncio, Multi-Agent)

Jun 2025

Multi-Agent Systems · Async Automation

- Designed a 5-agent orchestration workflow to automate topic selection, research, and outline generation, coordinating task handoffs and shared agent state.
- Implemented concurrent agents over Python asyncio to generate 7 niche titles per theme, each with research summaries, sources, and article outline guides.
- Cut research-and-ideation turnaround by ~80% (5 days → 1 session) with compiled, referenced sources; adopted as the team's standard ideation tool.

Tag Atlas (Next.js, Supabase, Gemini)

Personal project

Project write-up storage & tagging tool

- Built a self-hosted project-writeup archive, turning unstructured project history into a searchable, taggable board indexed by clickable tags instead of manual file hunting.
- Engineered a fuzzy markdown parser (Levenshtein-distance + token-similarity header matching) that converts arbitrary .md/.txt uploads into structured project fields, with heuristic detection that auto-splits multi-project files into individual drafts.
- Built a two-endpoint AI tagging system on the Gemini API using structured JSON prompting to recommend tags per project and cluster/merge near-duplicate tags board-wide.
- Implemented a live tag-based ranking algorithm that re-sorts projects client-side by selected-tag overlap on every click, with instant results and no server round-trip.

Leadership

Association for Computing Machinery (ACM) - BPHC Chapter

Nov 2024 - Jul 2026

Content & Research Team Co-Lead, Junior Office & Chair, Senior Office

- Authored 5+ technical articles for the ACM BPHC Blog and The BITSian Gazette, BITS Hyderabad's official magazine (Aug–Nov '24 edition).
- Conducted and coordinated logistics as Point of Contact for 4 tech events, including a coding workshop for 70+ indigent students.

English Language Activities' Society (ELAS), BPHC

Dec 2024 - Jan 2025

ARG Team Lead, Verba Maximus '25

- Led a 13-member team to design and execute a 6-hour event that drew 125+ participants - the highest-footfall headliner event at VM '25.

Talks & Workshops

Introduction to Machine Learning

Aug 2025

Organized by Association of Computing Machinery - BPHC Chapter

- Co-hosted a 250+ attendee workshop covering supervised/unsupervised learning, regression, neural networks, and activation functions with live coding.